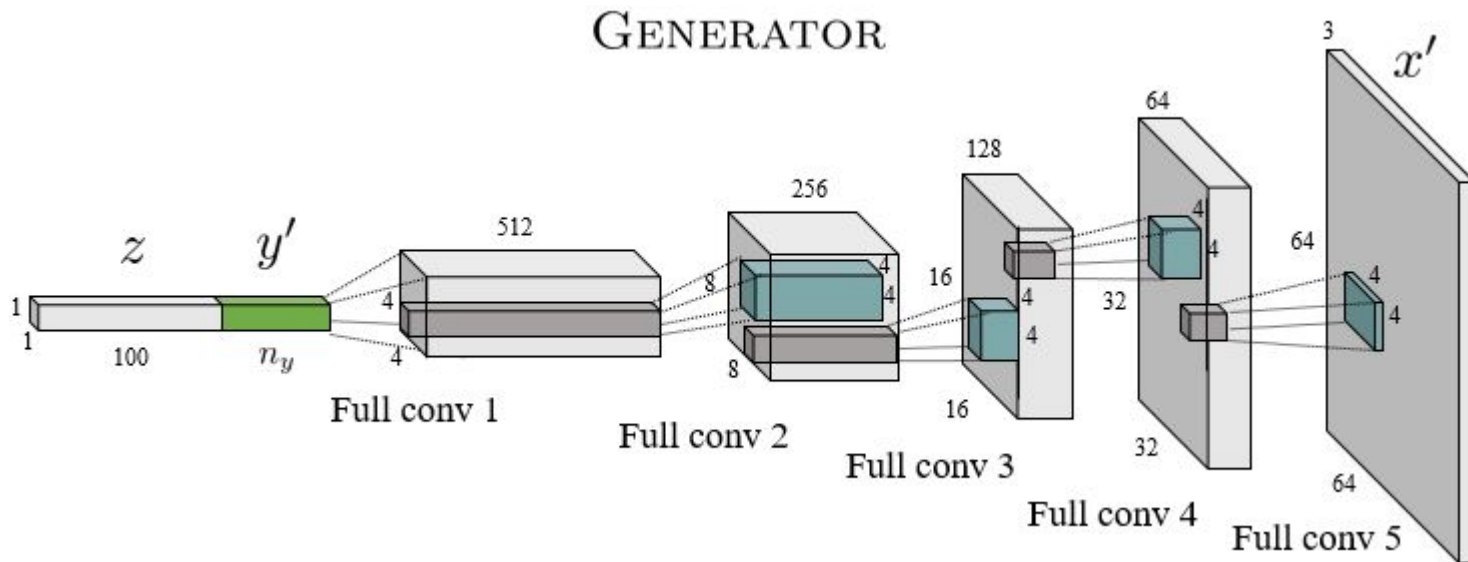


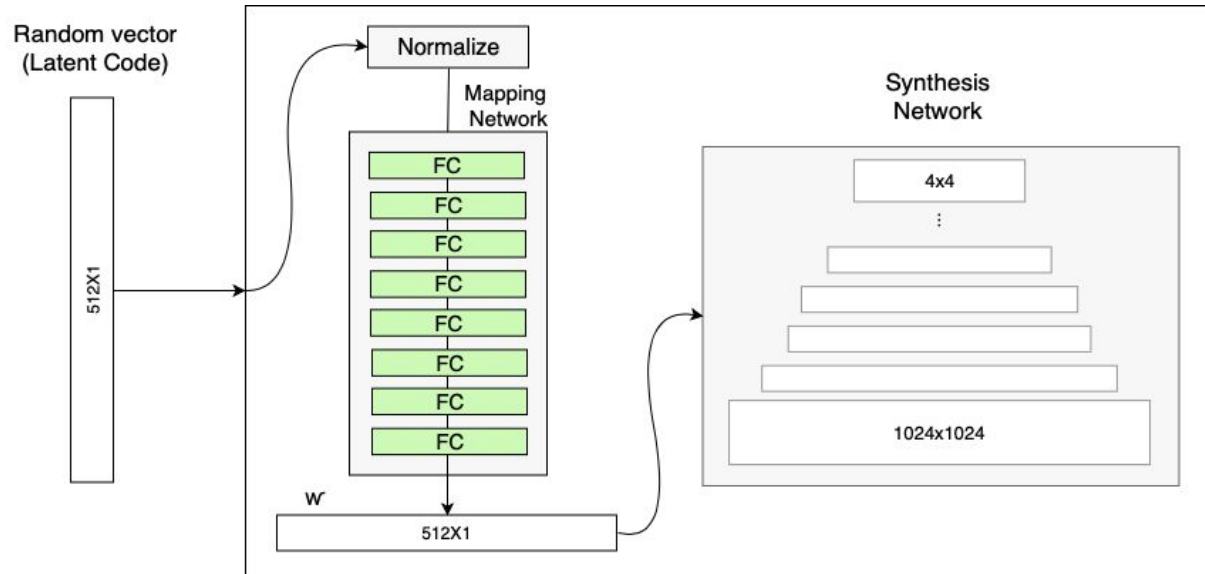


StyleGAN and application

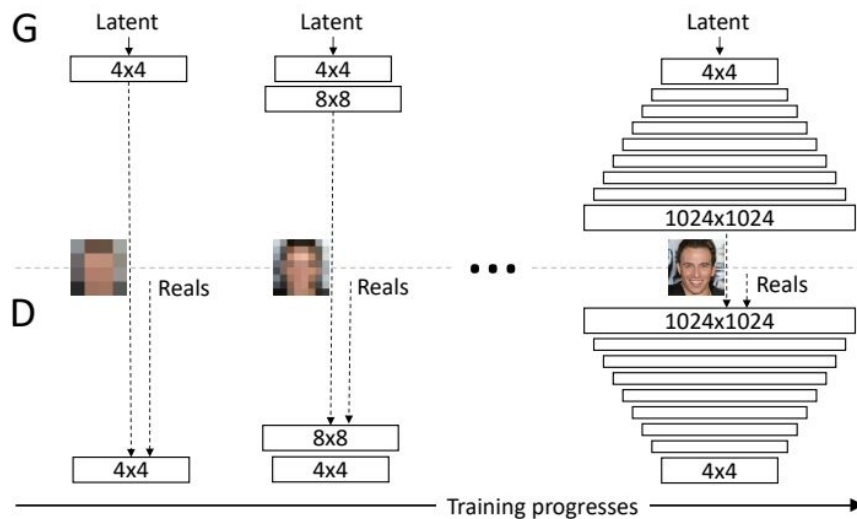
Feature entanglement in GAN



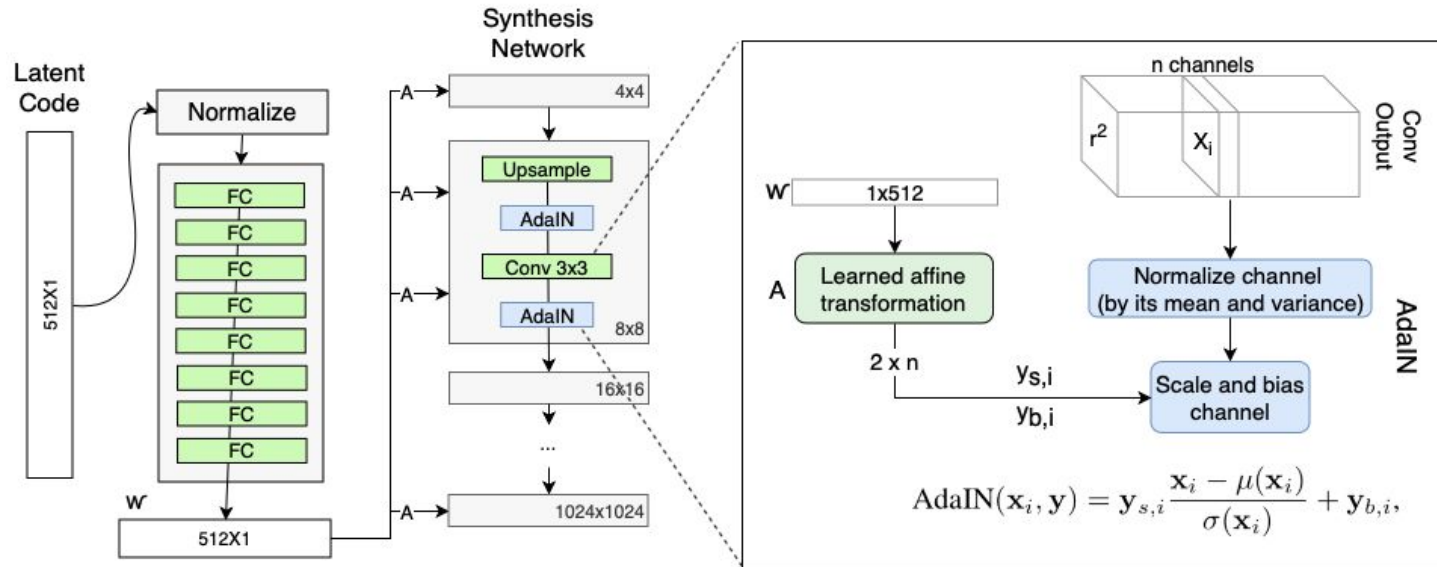
Network



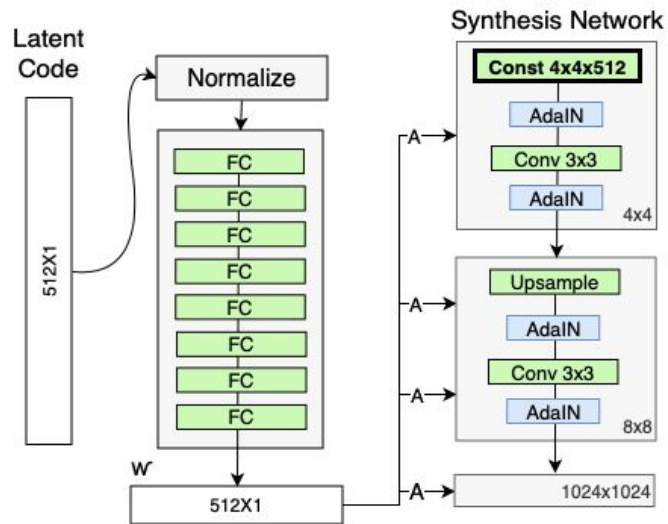
Synthesis network



Adaptive instance normalization



Remove noise input



1.



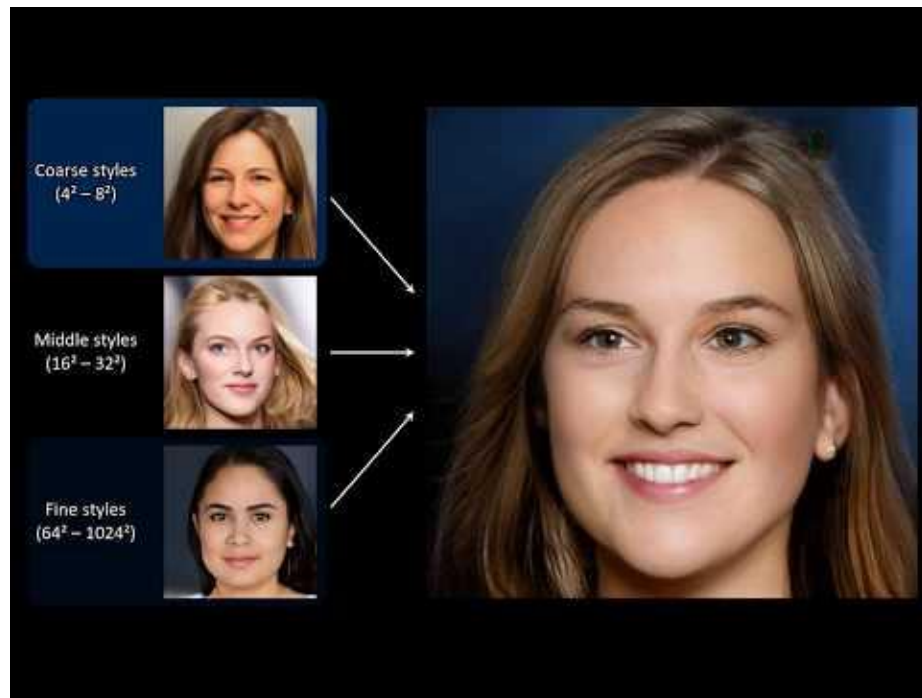
2.



Resolution structure

1. 4×4 - 8×8 : Pose, hair style, face shape
2. 16×16 - 32×32 : Finer facial features such as noise style, eye closed/opened
3. 64×64 - 128×128 : Color scheme (eye, hair, skin) and micro features

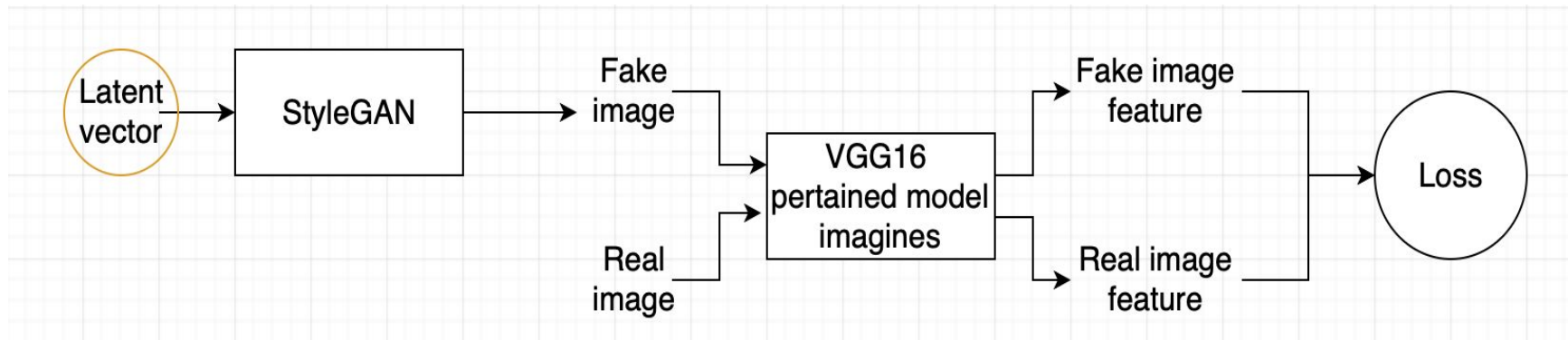
Style mixing



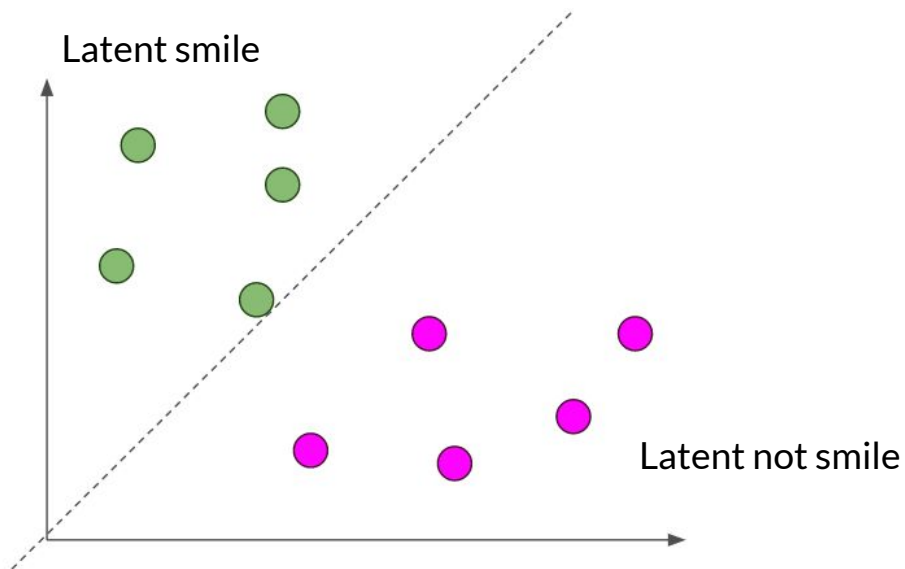
Result



StyleGAN encoder



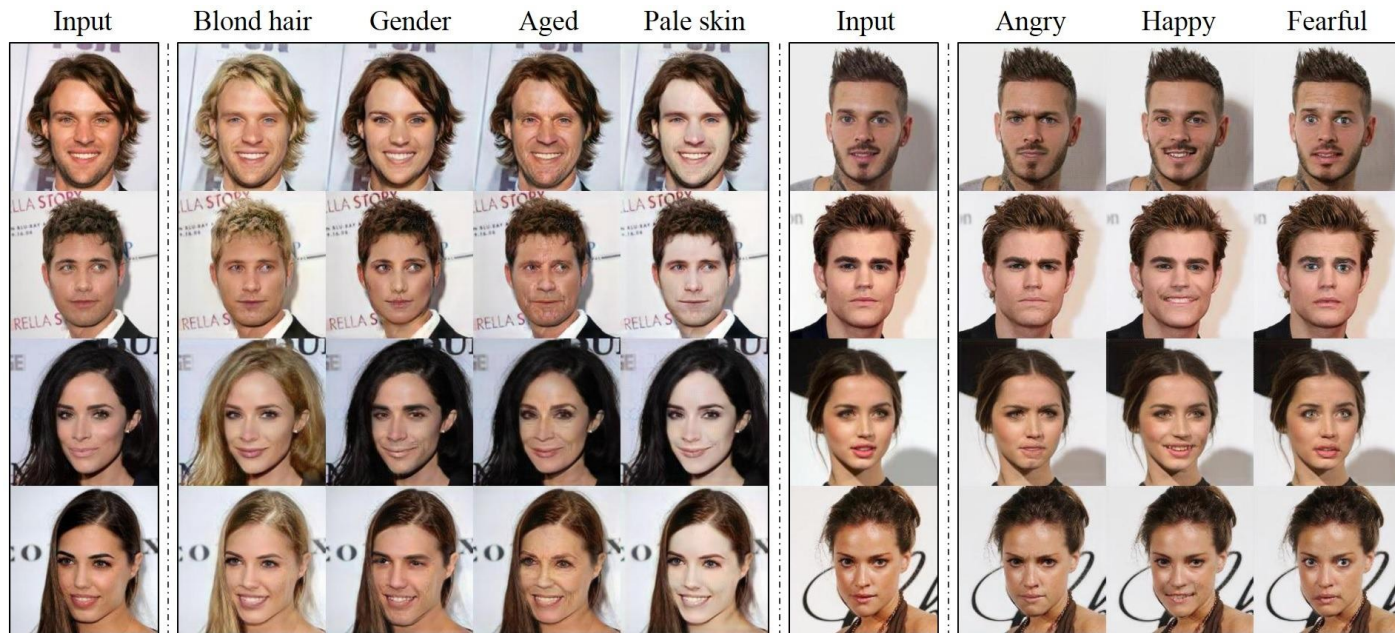
Find feature direction



Change image style



Stargan





Researching

1. Change the nationality of face in the image?
 - a. Stargan with each feature is nationality
 - b. Stylegan, get latent feature, then find nationality direction
2. Mix faces by using latent features

